
Quiet Orbit QO-R1: Design Reference

copperhead

<https://github.com/copperheadhq/copperhead-quiet-orbit>

A USB-C powered four-LED ambient lamp board, 57×57 mm

Abstract

Quiet Orbit QO-R1 is a desk ornament board: four warm-amber LEDs at the corners of a 40 mm square, each driven by its own hardware PWM output of an ATtiny84A and fading a quarter cycle apart, powered from USB-C at 5 V. The board is 57×57 mm, two layers, with all parts on the top side. This report describes the requirements, circuit, pin assignments, bill of materials, power budget, firmware, PCB, fabrication outputs and verification status of the design as committed. The schematic passes ERC and the board passes DRC with no exclusions. No board has been fabricated, the firmware has not been compiled, and no part has been checked against a current datasheet; one fitted part (D1) is known to be the wrong choice. Where this report and the source files disagree, the source files are correct.

Status: not built, not tested.

- No board has been fabricated. Nothing has been powered on.
 - The firmware has never been compiled, because no AVR toolchain was available where it was written.
 - Every manufacturer part number is unverified.
 - D1 (1N4148W) is the wrong part for the power path. Fit a 1 A Schottky (B5819W or 1N5819HW class) in the same SOD-123 footprint.
 - ERC and DRC pass. That shows the files agree with each other and with the design rules. It does not show that the circuit works.
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1 Product

A desk ornament powered from USB-C. Four warm-amber LEDs sit at the corners of a 40 mm square and fade independently, each a quarter cycle behind the last. The board has no USB data, no Power Delivery, no radio, no battery and no user controls. The firmware is written once through an AVR ISP header.

Board	57×57 mm, square, 2 copper layers, 1.6 mm FR-4
Assembly	All parts on the top side; SMT except the ISP header
Enclosure	80×80×25 mm with four light windows over the LEDs
Supply	USB VBUS, 5 V only
Controller	Microchip ATtiny84A, SOIC-14, internal 8 MHz RC oscillator
Outputs	4 amber LEDs, 590 to 605 nm, each on its own hardware PWM channel

Table 1: Product summary.

Out of scope: enclosure CAD, USB data, USB Power Delivery, wireless, batteries, user controls and compliance testing.

2 Requirements

Requirement identifiers match `docs/SPEC.md`. Items marked *assumed* are defaults chosen where the brief gave no value.

QO-001 USB-C power input

- USB-C receptacle, 5 V sink only.
- CC1 and CC2 each have their own 5.1 k Ω pull-down to GND.
- D+ and D– are not connected. No PD controller or other negotiation circuit.
- Receptacle centred on one board edge and flush with it.

QO-002 Input protection and filtering

- Resettable fuse sized for the load. *Assumed:* 100 mA hold, at least 200 mA trip at 25°C.
- Reverse-current protection so a programmer or header voltage cannot back-feed VBUS. *Assumed:* at most 10 μ A reverse leakage at room temperature with VCC at 5 V and VBUS unpowered.
- *Assumed:* at most 10 μ F nominal on the receptacle side; at least 1 μ F bulk after the protection; 100 nF at the MCU supply pins.

QO-003 Controller

- ATtiny84A in SOIC-14, powered directly from protected 5 V.
- Internal 8 MHz RC oscillator. No crystal, no external oscillator, no regulator.

QO-004 Lights and drive

- Four discrete amber LEDs, 590 to 605 nm, 1206 or a similarly hand-reworkable package.
- Each LED has its own series resistor and its own MCU pin with a hardware PWM output, so all four can fade independently at the same time.
- Full-brightness current 2 to 5 mA per LED. *Assumed:* 4 mA nominal target.
- LED pins may be shared with ISP signals only if programming still works with all LEDs and resistors fitted.

QO-005 LED placement

- Viewed from the component side, the LEDs are NW, NE, SE and SW, at the corners of a 40×40 mm square centred on the board. Optical centres, not footprint origins, set the position.

QO-006 Factory programming

- Standard 2×3, 2.54 mm AVR ISP header with pin 1 marked on the silkscreen.

- 10 k Ω pull-up on RESET.
- The board is powered from USB while programming. The ISP VCC pin is a voltage-sense connection only.

QO-007 Test access

- Separate test points for VBUS, VCC, GND and RESET, each labelled with its net name. *Assumed:* pad at least 1.0 mm across with 0.5 mm clear around it.

QO-008 Mechanical

- Square board, no larger than 72 $\times$ 72 mm. The committed board is 57 $\times$ 57 mm; the specification's original 70 mm starting outline was later reduced.
- Four M3 non-plated holes, one per corner, centres 4 mm from both adjacent edges. *Assumed:* 3.2 mm drill.
- All components on the top side.

QO-009 Marking

- Top silkscreen: "QUIET ORBIT / QO-R1", "USB-C 5V ONLY" near the receptacle, ISP pin 1, and the four test point names.

QO-010 Fabrication, assembly and sourcing

- Two-layer, 1.6 mm board built to standard fab rules.
- Total USB current strictly below 100 mA in every mode.
- All parts orderable from major distributors; SMT parts assemblable on the top side; the ISP header may be through-hole.

3 Circuit

The full schematic is reproduced in Appendix B and in `docs/schematic.pdf`.

3.1 Power entry

J1 is a 16-pin USB-C receptacle (GCT USB4105). All four VBUS contacts are joined. R1 and R2 (5.1 k Ω) pull CC1 and CC2 to GND, so a source sees a sink in either plug orientation and supplies default 5 V. D+, D–, SBU1 and SBU2 are unconnected. The shell is tied to board GND.

VBUS passes C1 (1 μ F, receptacle side), then F1 (100 mA PTC), then D1 (series diode, anode on the fuse side) to form VCC. C2 (4.7 μ F) is the bulk capacitor on VCC; C3 (100 nF) sits at the MCU supply pins. D1 blocks current from VCC back into VBUS, for example from a programmer that drives its VCC pin. The fitted 1N4148W drops up to about 0.8 V at 20 mA; Section 12 recommends a Schottky instead.

3.2 MCU

U1 is an ATtiny84A running from VCC with no regulator, on its internal 8 MHz RC oscillator. XTAL1 and XTAL2 (PB0, PB1) and PA0 to PA3 are unused. RESET has a 10 k Ω pull-up (R3) to VCC.

3.3 LED drive

Each channel runs from VCC through a 680 Ω resistor to the LED anode; the cathode goes straight to an MCU pin. The LEDs are active-low: driving the pin low turns the LED on. The four pins are the ATtiny84A's four hardware PWM outputs, split across Timer 0 (OC0A, OC0B) and Timer 1 (OC1A, OC1B).

With VBUS from 4.75 to 5.25 V, 0.5 to 0.8 V across D1, 1.8 to 2.4 V across the LED and 1% resistors, the current per LED is

$$I_{\text{LED}} = \frac{V_{\text{BUS}} - V_{\text{D1}} - V_{\text{F}}}{R} \in [2.26, 4.38] \text{ mA}, \quad (1)$$

inside the 2 to 5 mA requirement. The voltage limits are provisional; repeat the calculation with the fitted LED's datasheet values. A Schottky at D1 raises VCC by roughly 0.3 to 0.5 V and raises LED current slightly, so recheck the upper limit after that change.

3.4 Programming

J2 is a shrouded 2×3 AVR ISP header. SCK has its own pin (PA4). MOSI and MISO share pins with the SE and SW LED channels (PA6, PA5). The LED and its 680 Ω resistor load those lines while programming, so the SE and SW LEDs will flicker. A programmer with series resistance on MOSI may not pull the line far enough below the input-low threshold; a 1 kΩ LED resistor gives more margin. This has not been tested.

3.5 Test points

TP1 VBUS, TP2 VCC, TP3 GND and TP4 RESET are 1.5 mm round pads with their net names on the silkscreen.

4 Pin assignments

Pin	Port / function	Net	Use
1	VCC	VCC	Supply
2	PB0 / XTAL1	none	Unused (internal clock)
3	PB1 / XTAL2	none	Unused (internal clock)
4	PB3 / RESET	RESET	Reset, 10 kΩ pull-up, ISP
5	PB2 / OC0A	LED_NW	NW LED, PWM
6	PA7 / OC0B	LED_NE	NE LED, PWM
7	PA6 / MOSI / OC1A	ISP_MO_SE	SE LED, PWM; ISP MOSI
8	PA5 / MISO / OC1B	ISP_MI_SW	SW LED, PWM; ISP MISO
9	PA4 / SCK	ISP_SCK	ISP clock only
10 to 13	PA3, PA2, PA1, PA0/AREF	none	Unused
14	GND	GND	Ground

Table 2: U1, ATtiny84A.

Pin	Signal	Net
1	MISO	ISP_MI_SW
2	VCC (sense)	VCC
3	SCK	ISP_SCK
4	MOSI	ISP_MO_SE
5	RESET	RESET
6	GND	GND

(a) J2, AVR ISP.

Pins	Net
A4, A9, B4, B9	VBUS
A5	USB_CC1 (R1 to GND)
B5	USB_CC2 (R2 to GND)
A1, A12, B1, B12, shell	GND
A6 to A8, B6 to B8	not connected

(b) J1, USB-C.

Table 3: Connector pinouts.

Position	LED	Resistor	U1 pin	PWM	Shared ISP signal	Centre (mm)
NW	D2	R4	5	OC0A	none	(8.5, 8.5)
NE	D3	R5	6	OC0B	none	(48.5, 8.5)
SE	D4	R6	7	OC1A	MOSI	(48.5, 48.5)
SW	D5	R7	8	OC1B	MISO	(8.5, 48.5)

Table 4: LED channels. Centres are measured from the board's top-left corner, component side, *y* downwards.

5 Bill of materials

Every MPN is **unverified**. LCSC codes were matched from a third-party catalogue mirror on 22 September 2026 and have not been checked against a distributor. The same BOM is in `fab/2026-09-21-UNVERIFIED/` as a JLCPCB CSV.

Ref	Qty	Value	Footprint	MPN	LCSC	What to verify
J1	1	USB-C	GCT USB4105, 16P	USB4105-GF-A	C3020560	Pinout, shell, edge-flush geometry, rating
F1	1	100 mA PTC	0603	MF-FSMF010X-2	C210355	Hold and trip current, voltage, resistance, derating
D1	1	1N4148W	SOD-123	1N4148W-7-F	C81598	Replace with 1 A Schottky , then check reverse leakage ($\leq 10 \mu\text{A}$ at 5 V) and forward drop
C1	1	1 μF X7R	0603	CC0603KRX7R9BB105	none	Capacitance after DC bias, rating
C2	1	4.7 μF X7R, $\geq 10 \text{ V}$	0805	GRM21BR71A475KA73L	none	More than 1 μF retained at 5 V bias
C3	1	100 nF X7R	0603	CC0603KRX7R9BB104	C14663	Rating, placement at U1
R1, R2	2	5.1 k Ω 1%	0603	RC0603FR-075K1L	C105580	Value, tolerance
R3	1	10 k Ω 1%	0603	RC0603FR-0710KL	C98220	Value, tolerance
R4 to R7	4	680 Ω 1%	0603	RC0603FR-07680RL	C112310	Recalculate LED current from verified limits
D2 to D5	4	Amber LED	1206	150120AS75000	C5478408	Wavelength 590 to 605 nm, forward voltage at 2 to 5 mA, polarity
U1	1	ATtiny84A	SOIC-14, 3.9 mm	ATTINY84A-SSUR	C220597	Supply current, lifecycle; tube part -SSU is C144333
J2	1	AVR ISP 2 $\times$ 3	IDC 2.54 mm, vertical, THT	61200621621	none	Keying, pin 1, height
TP1 to TP4	4	Test point	1.5 mm pad	none	n/a	Bare copper; nothing to buy

Table 5: Bill of materials. 27 footprints in total: the 23 parts listed plus 4 mounting holes.

6 Power budget

Load	Worst case	Basis
CC terminations, R1 + R2	2.06 mA	5.25 V across 5.1 k Ω – 1%, twice
ATtiny84A	10.00 mA	Reserved, pending datasheet confirmation
Passive leakage and margin	1.00 mA	Allowance
Four LEDs	17.52 mA	$4 \times 4.38 \text{ mA}$
Total	30.58 mA	Design ceiling 40 mA; hard limit <100 mA

Table 6: Steady-state supply current. With RESET held low, R3 draws up to about 0.5 mA more.

7 Firmware

AVR-GCC and avr-libc for an ATtiny84A at $F_{\text{CPU}} = 8000000 \text{ UL}$. Source in `firmware/`. **Never compiled.**

Timers	Timer 0: fast PWM, OC0A and OC0B non-inverting, clk/64. Timer 1: 8-bit fast PWM, OC1A and OC1B non-inverting, clk/64
PWM frequency	8 MHz/64/256 $\approx$ 488 Hz on all four channels
Resolution	8 bits; compare value is 255 – brightness because the LEDs are active-low
Fade	Triangle wave from an 8-bit phase counter, advanced every 4 ms: $\approx 1.02 \text{ s}$ per cycle plus loop overhead
Offsets	NW 0, NE +64, SE +128, SW +192 (90° apart)
Build	<code>make -C firmware gives quiet-orbit-qo-r1.hex; flags -Wall -Wextra -Werror</code>
Checks	Compile-time error if F_{CPU} is not 8 MHz; all pin definitions are in <code>pins.h</code> only
Fuses	Default internal 8 MHz RC; clear CKDIV8 if the part ships with it set

Table 7: Firmware summary. Files: `main.c` (fade loop), `led_pwm.c/.h` (timers, brightness), `pins.h` (pin map from `docs/PINOUT.md`), `Makefile`.

8 PCB

8.1 Geometry

- 57×57 mm, 2 layers, 1.6 mm; all populated parts on the top side.
- LED optical centres on a 40 mm square centred on the board (Table 4). Each LED is vertical, cathode towards the horizontal centre line, resistor beside the anode.
- Four M3 holes, 3.2 mm NPTH, at (4, 4), (53, 4), (53, 53) and (4, 53) mm.
- USB-C centred on one edge and flush with it. C1, F1, D1 and C2 sit in a line directly behind it; C1 is 2.7 mm from the nearest VBUS pad.
- C3 is 2.4 mm from the MCU supply pin; R3 is 3.4 mm from RESET.

8.2 As committed

Footprints	27
Track segments	89
Track length	376.5 mm: 320.8 mm top, 55.8 mm bottom
Track widths	0.25, 0.3 and 0.5 mm
Vias	24, 0.65 mm pad, 0.3 mm drill; 21 GND stitching, one each on VCC, ISP_SCK and RESET
Copper pours	GND on both layers
Keepouts	A no-track area across the pad gap of every two-pad part (17)
Fiducials	None

Table 8: Board statistics, measured from the committed `.kicad_pcb`.

8.3 Design rules

Our own conservative rules, not a particular fab's.

Copper clearance	0.2 mm
Track width	0.25 mm signal, 0.5 mm power; minimum 0.2 mm
Via	0.65 mm diameter, 0.3 mm drill
Copper to edge	0.4 mm
Hole clearance	0.25 mm board-wide. One scoped exception: 0.19 mm between J1 and itself, because the stock USB4105 land pattern puts its GND pads 0.194 mm from its own locating-peg holes
Silkscreen text	At least 1.0 mm high
Severities	KiCad defaults, no exclusions

Table 9: Design rules.

8.4 Board size

With the LEDs on a 40 mm square and holes 4 mm from each edge, the LED courtyard meets the mounting-hole courtyard on the corner diagonal, which sets the minimum size.

Board edge	LED to hole courtyard gap	Note
56.0 mm	−0.22 mm	Overlaps; 6 courtyard errors
56.5 mm	+0.13 mm	Smallest that passes
57.0 mm	+0.47 mm	This board
58.0 mm	+1.17 mm	

Table 10: Courtyard clearance against board size.

57 mm is 32.5 cm², 34% less than a 70 mm board. Going smaller means moving the holes inside the LED square (about 46 mm) or also dropping the 40 mm LED pitch (about 31 mm); neither was laid out. Below 70 mm a single

board may not qualify for a fab’s factory-programming assembly tier without panelising; the ISP header is the fallback.

8.5 Layout notes

- The J2 ground pin and SMD ground pads connect solidly to the pours, without thermal relief; the ISP tracks leave no room for spokes. J2’s ground pin will be harder to hand-solder.
- J1 is surface-mount only on the board edge, with no mechanical stakes beyond its shell tabs.
- Tracks run under the “USB-C 5V ONLY” text. The text is on mask and stays legible.
- The longest nets are ISP MOSI at about 55 mm and VCC at about 117 mm.
- Routing came from an autorouter. Placement is fixed; rerouting gives a slightly different track count each time.

9 Milky Way variant

Same circuit and placement, with artwork on the back: the Milky Way over a mountain, made from copper with a mask opening (gold on ENIG), copper under mask (purple), bare laminate (dark purple) and silkscreen for the stars. Files are in `hardware/variant-milkyway/`; its `.kicad_pcb` and `.kicad_pro` replace the plain board’s.

- 4 mm rounded corners.
- No back GND pour. The artwork’s gold frame, including eyelets round the M3 holes, is the back GND ring instead.
- The artwork copper is unconnected polygons cut back from every pad, via, track and hole. A fab reviewer may query the floating exposed copper; mention it on the order.
- Needs ENIG. HASL would tin the exposed copper.
- DRC after the artwork was added, all severities with schematic parity: 0 violations, 0 unconnected, 0 parity issues.

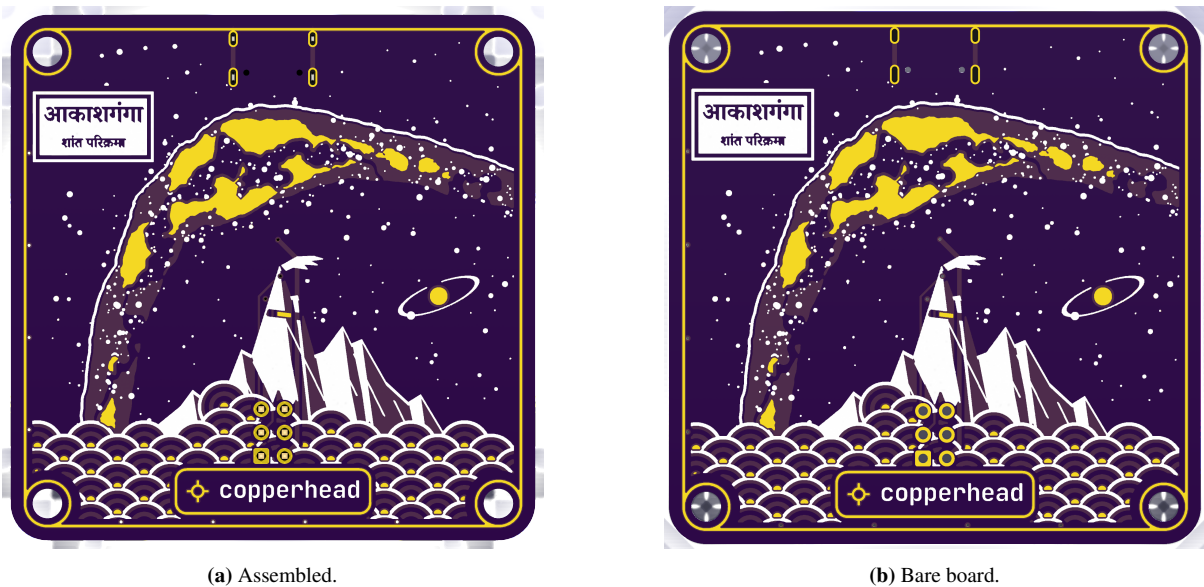


Figure 1: Milky Way back. KiCad renders.

10 Fabrication outputs

In `fab/2026-09-21-UNVERIFIED/`. Published so the design can be inspected; not a released fabrication package.

File	Contents
QO-R1-57mm-plain-jlcpcb.zip	Square board, GND pour on both sides
QO-R1-57mm-art-jlcpcb.zip	Milky Way back, 4 mm round corners, no back pour
plain/,art/	The same gerbers unzipped, plus drill maps and .gbr job
QO-R1-57mm-bom-jlcpcb.csv	BOM, transcribed, not verified; serves both boards
QO-R1-57mm-cpl-jlcpcb.csv	Placement, KiCad rotations, not corrected; serves both boards
plain-drc.json,art-drc.json	DRC reports taken just before plotting
make_bom_cpl.py	Generates the BOM and CPL

Table 11: Fabrication files.

Each zip	11 files: copper, mask, paste and silkscreen for both sides, Edge_Cuts, PTH and NPTH drill. Drill maps and job file left out so a fab cannot mistake a map for a layer
Gerbers	kicad-cli 10.0.4; Protel extensions, no X2, no netlist attributes, mask subtracted from silk, zones refilled, 4.6 format
Drill	Excellon, mm, decimal, absolute origin, PTH and NPTH separate, oval holes in alternate mode
Drill sizes	PTH 0.30 (vias), 0.60, 1.00 mm; NPTH 0.65 (connector pegs), 3.20 mm (M3)
Outline	Reads as 57.1 mm because the outline line has width; the board is 57×57 mm
Order notes	Plain: any colour, HASL or ENIG. Art: purple mask, ENIG. Neither has an order-number placeholder

Table 12: Plot settings.

11 Verification status

Check	Result
ERC (KiCad 10.0.4, all severities)	0 violations
DRC, plain board (all severities, schematic parity)	0 violations, 0 unconnected, 0 parity
DRC, Milky Way board	0 violations, 0 unconnected, 0 parity

Table 13: Checks run.

Not done: any physical build, power-on or measurement; compiling the firmware; checking any part against a current datasheet or distributor; a fab DFM check (the rules were compared with one fab's published 2-layer limits from memory only); opening the gerbers in a viewer; programming through the ISP header with the LEDs fitted.

12 Open items

Item	Action
D1 is a 1N4148W signal diode in the power path; it drops up to about 0.8 V.	Fit a 1 A Schottky (B5819W or 1N5819HW class) in the same SOD-123 footprint. No layout change. Recheck LED current.
BOM unverified; C1, C2 and J2 have no LCSC match.	Verify every line against a datasheet and a distributor.
Test points: DNP in docs/BOM.md, in-BOM in the schematic.	Decide which is right and make the other match.
ISP MOSI shares a pin with the SE LED through 680 Ω.	Test with the intended programmer; consider 1 kΩ for R6 and R7.
J1 GND pads 0.194 mm from its NPTH pegs.	Expect a query from fabs with a 0.2 mm limit.
CPL rotations uncorrected.	Check U1, J1 and the five diodes in the assembler's placement preview.
0.47 mm between LED and M3 hole courtyards.	Allow for the screw head in enclosure baffles; 58 to 60 mm gives more room.

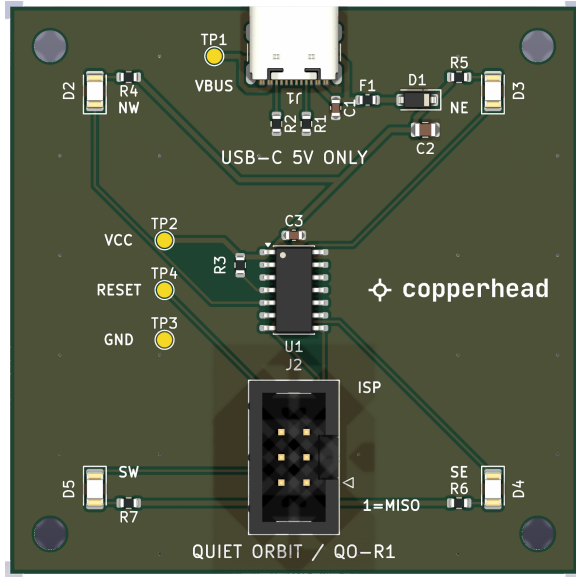
Table 14: Open items before ordering.

13 Licence

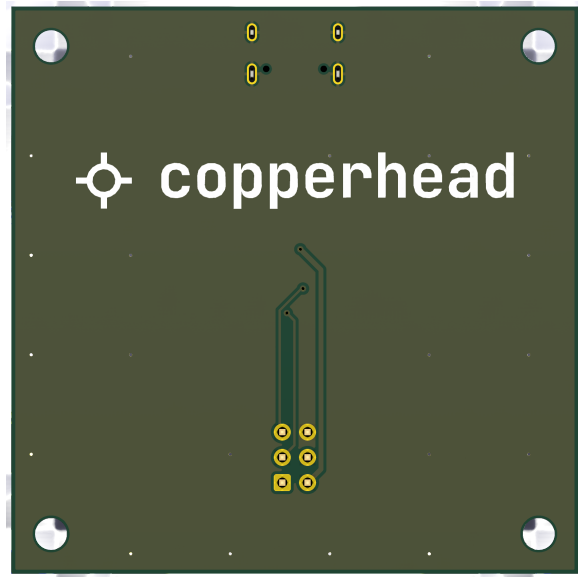
Hardware (`hardware/`, `fab/`, `3d/`) is CERN-OHL-S-2.0. Firmware (`firmware/`) is GPL-3.0-only. Documentation and images (`docs/`, `images/`, `README` files) are CC-BY-4.0. Footprints and symbols in `hardware/libs/` and `hardware/sym-lib-cache/` come from the KiCad libraries (CC-BY-SA 4.0 with the design-use exception). The copperhead mark on the silkscreen is a trademark and is not covered by these licences; remove it from modified boards. No warranty: this design has not been built or tested.

A Renders

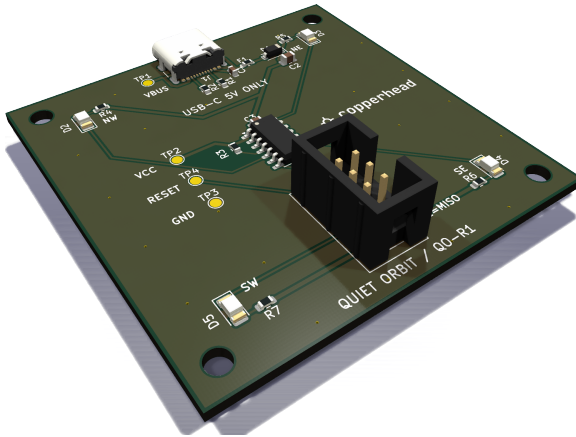
All images are KiCad raytraced renders with generic library 3D models. No board exists to photograph.



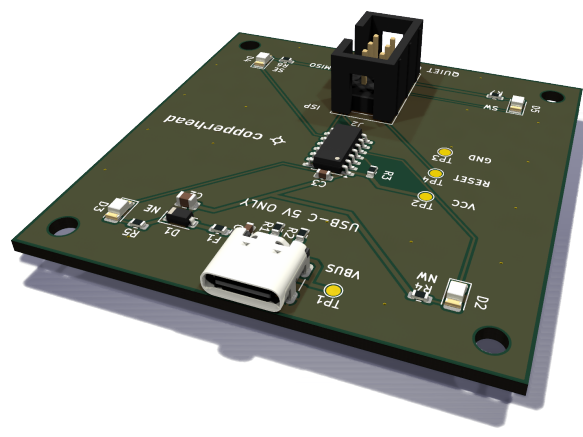
(a) Top, orthographic.



(b) Bottom, orthographic.

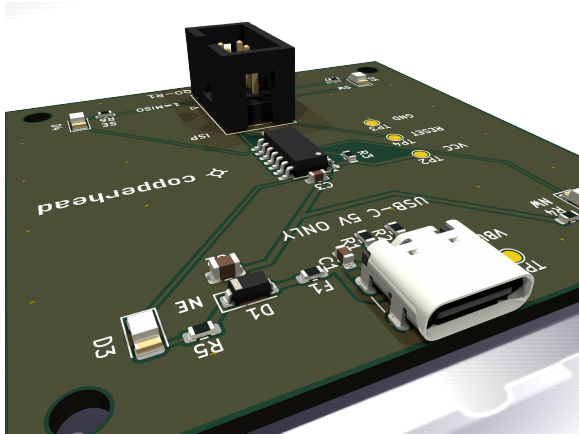


(c) From the ISP side.

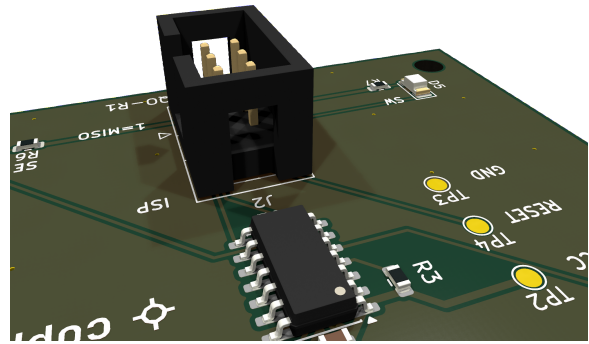


(d) From the USB side.

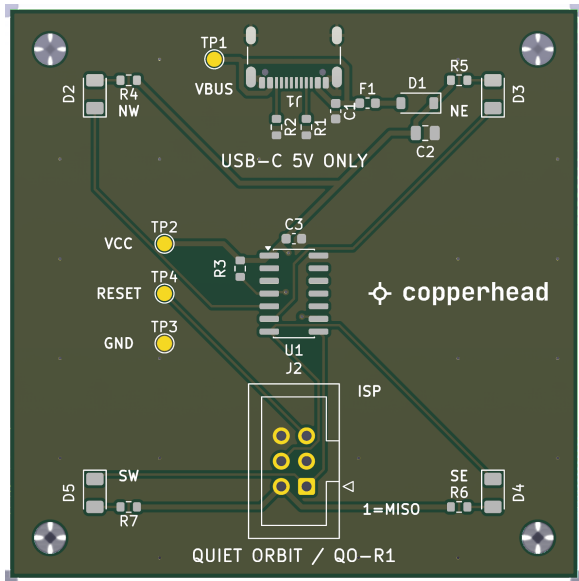
Figure 2: Assembled board.



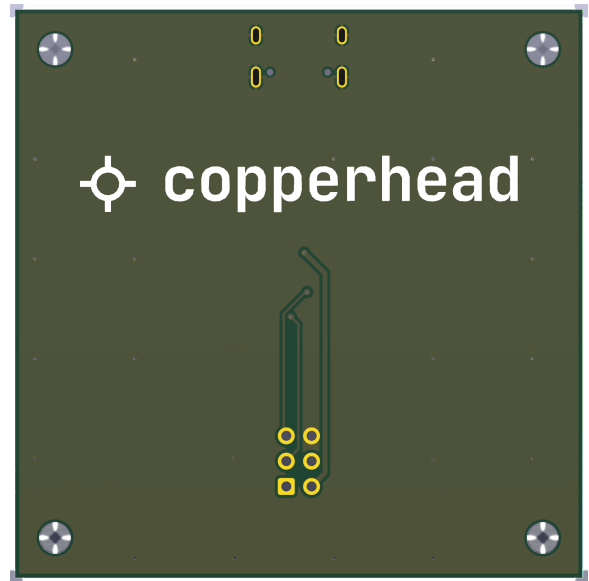
(a) Input chain: J1, C1, F1, D1, C2.



(b) U1, C3, R3 and J2.



(c) Bare board, top.



(d) Bare board, bottom.

Figure 3: Close-ups and bare board.

B Schematic

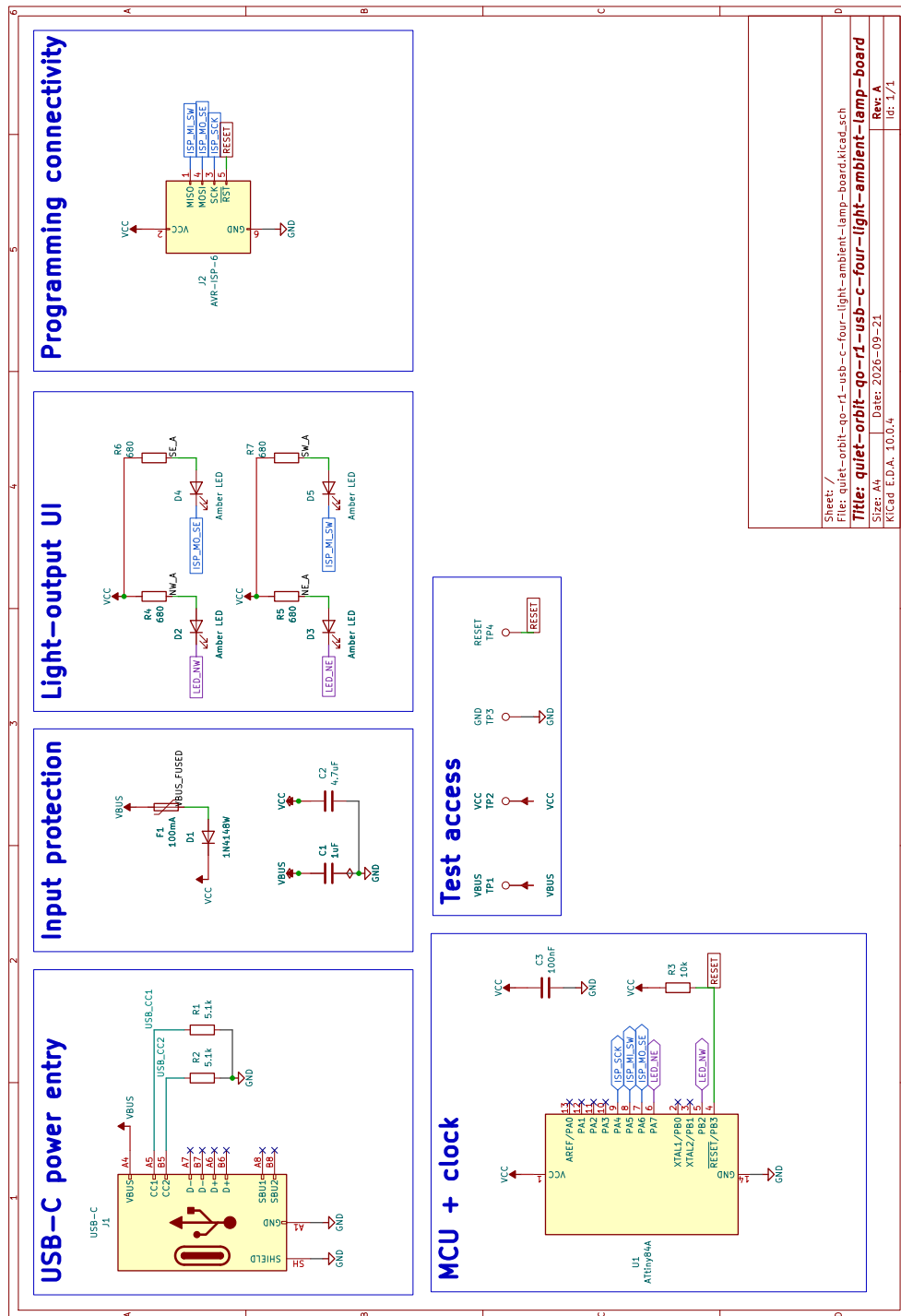


Figure 4: Full schematic sheet, rotated. Vector copies are in docs/schematic.pdf and docs/schematic.svg.